SAI SANTOSH CHEETHIRALA

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First-author paper submitting to TMLR, showing that the evidence responsible for a RAG error is usually plural rather than a single passage — which bounds every single-passage attribution method in use. Builds ML systems from first principles and takes them apart to learn why they fail.

EXPERIENCE

PESU Intelligence Labs — Research Intern, RAG & LLM Reliability **Bengaluru | Jun 2026 – Jul 2026**

No Single Culprit — Causal Attribution Benchmark & Evaluation Harness for Multi-Hop RAG **github.com/santoshcheethiralame-dot/LINEUP**

Python, PyTorch, HuggingFace Transformers, vLLM, Conformal Prediction, BM25, pytest, GitHub Actions (CI/CD), Streamlit

- Built a 7.2K-LOC, CI-tested evaluation harness (135 automated tests on GitHub Actions) that certifies which retrieved passages cause a multi-hop RAG (retrieval-augmented generation) error, with a public evaluation API, pooled leaderboard, and Streamlit explorer
- Designed a non-circular counterfactual oracle (leave-one-out roles + exhaustive minimal-sufficient-set enumeration) producing leak-free ground-truth labels without ever consulting the attribution method under test, ending circular evaluation
- Showed the responsible evidence is typically plural, disjoint, and non-monotone (52% no single sufficient passage, 28% two disjoint sufficient sets, 67-87% non-monotone), bounding single-passage attribution at a 1/m recall ceiling, then replaced it with a calibrated set-valued method: 94% conformal coverage, margin-based abstention (AUROC 0.77) lifting top-1 accuracy from 0.35 to 0.64
- Ran ~100K teacher-forced inferences across open-weight models (7B-120B) x three multi-hop datasets with a live BM25 retriever; verified the structure holds to deployment depth (k=20-50) and 120B scale, with preregistered hypotheses and reported nulls. First-author paper submitting to TMLR (Transactions on Machine Learning Research)

PROJECTS

DRAGNET — Distribution-Free Set-Valued Causal Attribution for RAG **github.com/santoshcheethiralame-dot/DRAGNET**

Python, split-conformal prediction, subset-lattice search, budgeted extraction

- Extended the benchmark into a deployable method: estimates the minimal sufficient causal set — the smallest passage subset that alone reproduces an answer — under an explicit model-query budget, wrapped in a distribution-free coverage guarantee
- Implemented exact minimal-sufficient/necessary-set search over the subset lattice as validation-only ground truth, plus a budgeted extractor and an order-2 subset surrogate that separates redundancy from joint support

GlassBox — Full-Stack Model Interpretability Platform (Deployed) **numero00-glassbox.hf.space**

Python, FastAPI, React, TypeScript, Docker, REST APIs, PyTorch, TransformerLens, SAELens

- Halved a Gemma hallucination's memorized-answer logit gap (-9.2 to -4.4) by ablating a single SAE feature, demonstrating feature-level causal control; sparse autoencoders resolve the residual stream into 24K+ interpretable features
- Cross-validated attribution patching against exact activation patching, so the cheap attribution method is checked against the expensive ground truth rather than trusted

Gradient — GPU-Native Deep-Learning Framework, In-Browser **santoshcheethiralame-dot.github.io/GRADIENT**

TypeScript, WebGPU, WGSL, React, Web Workers

- Built a from-scratch framework that trains neural nets entirely on the GPU in-browser (25+ hand-written WGSL kernels, hand-derived backprop, Adam, no autodiff), with no Python, server, or WASM in the loop
- Validated every gradient three ways (f64 CPU oracle 51/51, finite-difference, PyTorch parity to ~1e-7) and profiled a tiled matmul to ~3-4x a naive kernel

TECHNICAL SKILLS

ML & Research: PyTorch, HuggingFace Transformers, conformal prediction, causal attribution, mechanistic interpretability (sparse autoencoders, activation & attribution patching), RAG, graph neural networks (GraphSAGE, GAT), 4-bit quantization

Languages: Python, C, Rust, TypeScript, JavaScript, SQL, WGSL

Systems & Backend: Distributed systems (Raft, LSM storage, MVCC), FastAPI, REST APIs, Docker, Linux, WebGPU/WGSL, vLLM

Testing & Reliability: Deterministic simulation testing, randomized fault injection, linearizability checking, regression testing, pytest, CI/CD (GitHub Actions), model evaluation

Tools: Git, GitHub Actions, Linux, Hugging Face Spaces, GPU compute, numerical gradient checking

EDUCATION

PES University — B.Tech, Computer Science Engineering **Bengaluru | Sep 2024 – May 2028 (Expected)**

CGPA 7.95/10 (through Sem 4) — CNR Merit Scholarship (top 25% cohort), Sem 1 — Academic Distinction Award (SGPA 7.75+), Sem 2 & Sem 4

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, NLP, Linear Algebra, Data Science

LEADERSHIP & ACTIVITIES

Sahay AI — Student AI Club, PES University — Multimedia Domain Head **Oct 2025 – Mar 2026**

- Built a content pipeline and posting schedule for a 50+ member AI club, driving 5-6x growth in content views over three months
- National-Level Air Rifle Competitor - 10m Air Rifle, National School Games (a precision event built on sustained focus)